

Mahija Ibad Pradipta

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PERSONAL SUMMARY

Computer Engineering graduate from Sepuluh Nopember Institute of Technology (ITS) with strong expertise in full-stack software development, IT infrastructure, DevOps, and networking. Proven track record in building scalable web and mobile applications, designing containerized systems and homelab environments, and developing autonomous edge IoT/robotics solutions. Demonstrated leadership, technical mentoring, and problem-solving skills through national robotics championships, organizational leadership, and laboratory teaching assistantships.

EDUCATION

Sepuluh Nopember Institute of Technology — Surabaya <i>Bachelor of Computer Engineering</i>	July 2022 – August 2026 GPA: 3.32/4.00
SMAN 4 Jakarta — Jakarta <i>Science Major</i>	July 2019 – May 2022 Final Score: 86/100

WORK EXPERIENCE

Digital Signal Processing Laboratory Course — M-IOT ITS <i>Staff - Teaching Assistant</i>	October 2025 – December 2025 On-site
- Assisted and supervised students during Digital Signal Processing laboratory sessions - Evaluated students' understanding of course materials, assignment completion, and discipline throughout laboratory activities - Reviewed and assessed Digital Signal Processing laboratory reports	
Introduction to Programming Laboratory Course — M-IOT ITS <i>Secretary and Treasurer - Teaching Assistant</i>	July 2025 – December 2025 On-site
- Supervised, guided, and evaluated students during laboratory sessions, mentoring activities, laboratory reports, and final projects - Prepared laboratory modules and managed administrative activities - Managed attendance records and assessment recapitulation for teaching assistants and students	
PT Winnicode Garuda Indonesia — Bandung <i>Intern - Full-Stack Developer</i>	March 2025 – September 2025 Remote
- Developed UNTERN (U Need Intern), a web application connecting internship seekers with companies seeking talent - Managed the entire application development lifecycle, including design, implementation, testing, and deployment - Analyzed user requirements and translated them into application features and functionalities	
PT Lintasarta — Central Jakarta <i>Intern - IT Services Management (Infrastructure)</i>	July 2025 – August 2025 On-site
- Supported activities within the IT Infrastructure subdivision under the IT Services Management Department, Corporate IT Division - Participated in hardware deployment processes to support company operations - Learned enterprise network architecture and IT service management practices within a corporate environment	
Introduction to Programming Laboratory Course — M-IOT ITS <i>Staff - Teaching Assistant</i>	September 2024 – March 2025 On-site
- Assisted Electrical Engineering, Telecommunications Engineering, and Biomedical Engineering students during Introduction to Programming laboratory sessions - Guided students through Introduction to Programming mentoring activities - Evaluated final project presentations and laboratory reports	
Robotics Extracurricular — SDN Klampis Ngasem I/246 - Surabaya <i>Instructor</i>	August 2023 – June 2024 On-site
- Guided and mentored 4th, 5th, and 6th grade students participating in robotics extracurricular activities - Taught fundamental robotics concepts through theory and hands-on practice using the 4-Wheel Obstacle Avoidance Robot Kit - Delivered robotics learning materials through interactive web-based simulations	

PROJECTS

- Job Tracker Monorepo & Automated Messaging Gateway** | *NestJS, Next.js, TypeScript, PostgreSQL, Prisma, Redis, Docker* 2026
- Architected a production-grade recruitment tracking monorepo via Turborepo featuring NestJS REST API, Next.js UI, RBAC security, and Socket.IO events
 - Integrated Google Gmail API with intelligent keyword filtering and parsing logic to automatically extract and synchronize recruitment email updates
 - Built bidirectional WhatsApp and Telegram bot notification gateways, with multi-stage Alpine Docker builds (< 180 MB) and automated CI/CD pipelines
- Finance Tracker & Observability Engine** | *NestJS, TypeScript, Prisma, PostgreSQL, Redis, MinIO S3, OpenTelemetry* 2026
- Engineered a multi-account financial tracking ledger supporting cash, bank, e-wallet, and investment accounts with budget threshold alert notifications
 - Developed automated WhatsApp bot transaction logging with MinIO S3 receipt object storage and BullMQ asynchronous background queue processing
 - Implemented OpenTelemetry distributed tracing (Jaeger) and Prometheus metrics (prom-client) for full-stack observability and APM monitoring
- DIY Smart CCTV Monitoring System** | *Docker, EMQX, MinIO, MySQL, Laravel, Webmin, Cloudflared* 2026
- Architected and deployed a self-hosted CCTV platform on an Intel NUC server integrated with ESP32-CAM edge devices for image collection and monitoring
 - Managed a multi-service Docker environment including Laravel, MySQL, MinIO, phpMyAdmin, EMQX MQTT broker, and supporting application services
 - Implemented MQTT-based data communication, server administration and monitoring through Webmin, and secure public access using Cloudflare Tunnel
- Autonomous Human Search System Using Drone with Pi 5** | *YOLO, ONNX, MAVSDK, Edge Device* 2026
- Designed and integrated a drone power distribution system capable of providing stable power for Raspberry Pi 5 computing requirements
 - Evaluated and optimized YOLOv8n ONNX model performance in a headless Linux environment on Raspberry Pi 5 for real-time human detection
 - Developed an autonomous MAVSDK-based system with autonomous navigation, victim documentation, and duplicate target tracking prevention capabilities
- Self-Hosted Homelab Infrastructure** | *Debian, Docker, Cloudflare Tunnel, Webmin, Linux Administration* 2025
- Built and maintained a self-hosted homelab environment using repurposed laptop hardware to run personal applications and services
 - Deployed and managed multiple web applications and supporting services using Docker containers on a Debian-based server
 - Implemented secure remote access through Cloudflare Tunnel and performed server administration and monitoring using Webmin
- SwiMate** | *Flutter, Dart, Supabase, PostgreSQL, TensorFlow Lite* 2025
- Developed a mobile application to support swimming training and performance monitoring
 - Integrated mobile sensors and machine learning models to support activity analysis
 - Implemented backend services and data management using Supabase and PostgreSQL
- UNTERN** | *JavaScript, PostgreSQL, Vite* 2025
- Developed a web platform connecting internship seekers with companies seeking talent
 - Managed the entire software development lifecycle from requirements analysis to deployment
 - Implemented application features and system functionalities to support the internship recruitment process
- SARVIO-X** | *JavaScript, DJI Tello, PyTorch* 2025
- Developed a web-based application for monitoring and interacting with DJI Tello drones
 - Implemented user interface components and drone control functionalities
 - Supported drone operations and monitoring through a browser-based platform
- Course Web Application** | *JavaScript, PostgreSQL* 2025

- Developed a web application for course management and learning activities
- Implemented frontend and backend functionalities to support academic workflows
- Designed features for managing course-related information and user interactions

CarVole 2D Car Game | *C++, graphics.h* 2024

- Developed a 2D car game using C++ and graphics.h
- Implemented game mechanics, object movement, and collision handling
- Designed interactive gameplay elements and user controls

GUI Application with wxWidgets | *C++, wxWidgets* 2023

- Developed a desktop graphical user interface application using wxWidgets
- Applied object-oriented programming principles in application development
- Implemented interactive user interface components and application features

AWARDS

With Banyubramanta ITS:

5th Place - Singapore Autonomous Underwater Vehicle Challenge (SAUVC) 2025 March 2025

1st Place - Indonesian Underwater Robot Competition National Level 2024 July 2024

3rd Place - Indonesian Underwater Robot Competition Regional Level II 2024 June 2024

TRAINING

Cisco Certified Network Associate (CCNA) Training — ID-Networkers August 2026

Intermediate Student Management Skills Training (LKMM-TM) August 2024

Basic Student Management Skills Training (LKMM-TD) November 2023

Pre-Basic Student Management Skills Training September 2023

MAGE 9 Multimedia Workshop August 2023

Basic Student Entrepreneurship Skills Training (LKMW-TD) November 2022

Basic Scientific Writing Training (PKTI-TD) October 2022

ORGANIZATIONS

M-IOT Laboratory — Computer Engineering ITS — Surabaya June 2025 – May 2026
Coordinator - Board of Directors *On-site*

- Improved laboratory quality in both project development and human resource development aspects
- Supervised overall timelines and task execution across all divisions and subdivisions
- Coordinated laboratory management activities with external stakeholders and university administration

HIMATEKKOM ITS — Surabaya March 2025 – February 2026
Head of Cadre Development Department - Board of Directors *On-site*

- Guided IC and OC teams in executing all cadre development activities
- Mentored SC members in designing assessment systems and activity timelines
- Coordinated with students and the Computer Engineering Department regarding cadre development activities

M-IOT Laboratory — Computer Engineering ITS — Surabaya August 2024 – June 2025
Archive Staff - Internal Division *On-site*

- Managed archives of thesis topics and telematics projects conducted by M-IOT Laboratory assistants
- Maintained archives of midterm and final examination materials for M-IOT specialization courses
- Compiled learning materials for M-IOT specialization courses

Banyubramanta Robotics Team ITS — Surabaya July 2024 – May 2025
Head of Official Division - Board of Directors *On-site*

- Led the Administration & Finance, Public Relations & Sponsorship, and Creative & Branding subdivisions
- Planned and supervised overall timelines and task execution across all subdivisions
- Ensured smooth execution of the team's administrative and non-technical operations

HIMATEKKOM ITS — Surabaya

March 2024 – March 2025

*Secretary and Treasurer Staff - Board of Directors**On-site*

- Coordinated with the Computer Engineering Department regarding organizational administration
- Prepared outgoing letters, meeting minutes, and certificate submissions for organizational activities
- Maintained archives of important documents in both digital and physical formats

HIMATEKKOM ITS — Surabaya

April 2024 – December 2024

*Steering Committee Staff - Cadre Development Department**On-site*

- Coordinated with IC (Instructor Committee) and OC (Organizing Committee) regarding cadre development activities
- Designed student assessment systems and overall cadre development timelines
- Coordinated with students and the Computer Engineering Department regarding cadre development activities

Banyubramanta Robotics Team ITS — Surabaya

February 2023 – July 2024

*Senior Finance Staff - Official Division**On-site*

- Recorded team income and expenditures
- Managed team budget allocation
- Prepared accountability reports, financial reports, and budget plans

Banyubramanta Robotics Team ITS — Surabaya

November 2022 – January 2023

*Finance and Administration Intern - Official Division**On-site*

- Prepared sponsorship and administrative proposals
- Prepared contracts, accountability reports, and budget plans
- Created content for the team's social media platforms

ITS Robotics Club — Surabaya

September 2022 – February 2023

*Member**On-site*

- Actively participated in organizational programs and activities

ITS IBC Club — Surabaya

September 2022 – September 2023

*Member**On-site*

- Actively participated in organizational programs and activities

COMMITTEES

Multimedia And Game Event 10 — Computer Engineering ITS

February 2024 – February 2025

*Head of Event Division - Board of Directors**On-site*

- Guided staff in designing and executing IoT, Multimedia, and Robotics workshops
- Guided staff in planning and executing opening ceremonies, talk shows, exhibitions, and closing events
- Supervised and evaluated the implementation of workshops, talk shows, exhibitions, opening ceremonies, and closing events

Multimedia And Game Event 9 — Computer Engineering ITS

February 2023 – March 2024

*Staff - Secretariat Division**On-site*

- Managed activity data collection and documentation
- Prepared communication materials, attendance records, feedback forms, and meeting minutes
- Served as the person in charge of one workshop

HGTS X SE 8.0 — HIMATEKKOM ITS

October 2023

*Staff - Teaching Division**On-site*

- Designed learning concepts for students of MTs 19 Surabaya
- Prepared presentation materials for teaching activities
- Taught image detection and pose detection topics to students of MTs 19 Surabaya

ISC - 63rd ITS Anniversary — ITS

September 2023 – November 2023

*Badminton Staff - Event Division**On-site*

- Ensured the smooth execution of the inter-faculty badminton tournament
- Coordinated match schedules and tournament facilities
- Managed documentation and match result records

TDC Summit Fest 2023 — TDC ITS

August 2023 – November 2023

TDC For Startup Staff - Event Division

On-site

- Prepared implementation and technical guidelines for startup bootcamp mentoring activities
- Prepared implementation and technical guidelines for startup bootcamp coaching activities
- Served as Master of Ceremony (MC) during startup bootcamp mentoring sessions

SPS 128 — HIMATEKKOM ITS

June 2023 – September 2023

Staff - Event Division

On-site

- Prepared implementation and technical guidelines for the 128th FTE ITS Graduation Appreciation Event
- Prepared event rundown for the 128th FTE ITS Graduation Appreciation Event
- Served as Liaison Officer for the Head of the Computer Engineering Department

Inclenation 2023 — BEM FTEIC ITS

June 2023 – August 2023

Staff - Mentor Division

On-site

- Mentored 20 freshmen from 6 departments and 10 study programs within FTEIC
- Provided guidance and support to freshmen throughout Inclenation 2023
- Served as an internalization facilitator during Inclenation 2023

SKILLS

- **Programming Languages:** JavaScript, Python, C++, Dart, SQL
- **Web Development:** Laravel, Vite, REST API, PostgreSQL, MySQL, React
- **Mobile Development:** Flutter
- **Infrastructure & DevOps:** Linux (Debian & Ubuntu), Docker, Cloudflare Tunnel, Webmin, MinIO, EMQX
- **Networking:** TCP/IP, MQTT, Network Troubleshooting, WebSocket
- **AI & Computer Vision:** YOLO, ONNX, TensorFlow Lite, OpenCV
- **Tools:** Git, GitHub, Postman, phpMyAdmin
- **Languages:** Indonesian (Native), Javanese (Conversational), English (Professional Working Proficiency)